# Submission metadata and declarations

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## Article title

Characterisation and Quantification of Data Centre Flexibility for Power System Support

## Article type

Full Length Research Article.

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## Keywords

Data centre; power system flexibility; demand side flexibility; load shifting; thermal inertia

## Previous submission

This is a substantially revised resubmission of APEN-D-25-21553, previously rejected by Applied Energy. The previous decision letter permits resubmission after significant revision. Responses address the reports of Reviewers 2 and 5. The previous submission listed Mehmet Türker Takci first; James Day is now first author.

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## Competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## CRediT authorship contribution statement

James Day: Writing – original draft, Investigation, Methodology, Conceptualization.
Mehmet Türker Takci: Writing – original draft, Investigation, Methodology, Conceptualization.
Meysam Qadrdan: Writing – review & editing, Supervision, Conceptualization.

## Generative AI declaration

During the preparation of this work, the authors used ChatGPT Codex (OpenAI) and Claude Code (Anthropic) to assist with manuscript preparation, editing and coding, including the Python code used to plot figures from the supplied numerical inputs and results, the workflow diagram and the redrawn

facility schematic, as identified in the figure captions. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

## Data availability

All input data used in this study are publicly available or reported in full in the article. The electricity prices are the publicly available Low Carbon Contracts Company IMRP actuals; the adopted workload profiles, deferral tranches and model parameters are given in Tables I, III and IV and Supplementary Section S1. The results supporting the findings are reported in the article and its supplementary information.

## Author confirmations

The corresponding author confirmed the author order, corresponding-author role, approval by all authors, originality/exclusive submission, no competing interests, funding completeness and funder role. No additional reviewer reports were received.

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## Abstract

Data centres can support power systems by coordinating workload scheduling, electrical storage and cooling, but their available flexibility depends on the operating state and required service duration. This paper develops an integrated model of IT workload shifting, uninterruptible power supply (UPS) batteries, cooling and thermal energy storage. Linked daily mixed-integer linear optimisations establish a cost-minimising operating baseline over 2025. A duration-aware assessment then identifies certified feasible power deviations for specified start times while enforcing workload, storage, thermal and post-event recovery constraints. For a hypothetical data centre with 1 MW of rated IT capacity, coordinated operation reduces annual electricity costs by GBP 29,626.91 (5.38%). Resource sensitivities show the largest change in savings when the flexible workload share is varied, followed by UPS and thermal storage energy capacity. On the selected representative day, a 100 kW load reduction is feasible for six hours from noon and one hour from 06:00. Morning reductions rely predominantly on UPS discharge, whereas afternoon reductions rely mainly on workload deferral and associated cooling reductions. Load increases combine workload execution, cooling and storage actions. The results quantify feasible magnitude–duration combinations under the stated model and recovery assumptions; assessment caps and unresolved trials do not establish maximum durations. Lower electricity costs do not imply lower annual electricity consumption or peak import. The framework provides candidate flexibility offers for subsequent assessment against network constraints and service-specific requirements.